MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

Artificial Intelligence and Society

Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 2.3: Bias and Variance

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Sources of Error

- In the section on performance indicators, we saw that there are **different types** of errors.
- There are also **different reasons** why machine learning models make errors.
- These reasons can lie in both the **training data** and the **training process**.

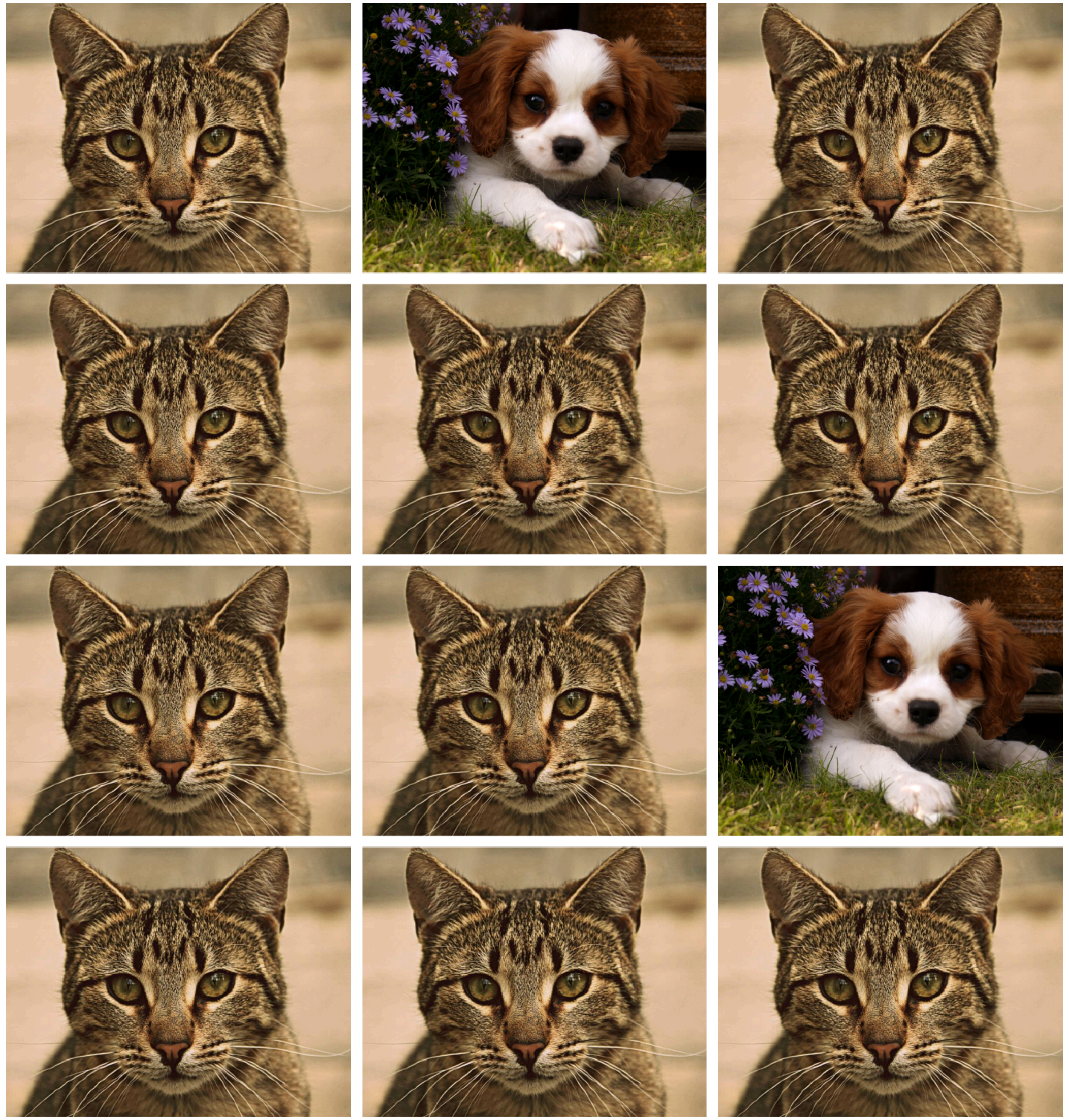
Next, we will look closer at these reasons.

Bias

- Bias of a model: Difference between its **average prediction** and the true value.
- Example
  - We train a model to recognize cats and dogs in images.
  - Cats appear much more frequently in the training dataset.
  - As a result, the model is much more likely to predict "cat" than "dog".
  - The model has a cat bias!

ethical aspects of AI

legal aspects of AI



Source cat image: Wikimedia Commons CC BY-SA.
Source dog image: Wikimedia Commons, CC BY.

Variance

- Variance of a model: Spread of predictions for similar input observations.
- **Example**
  - We train a model to recognize ponies and horses in images.
  - Only brown horses and white ponies appear in the training data.
  - The model memorizes the training data and learns to associate brown → horse and white → pony.
  - The performance on new data with white horses and brown ponies is extremely poor.

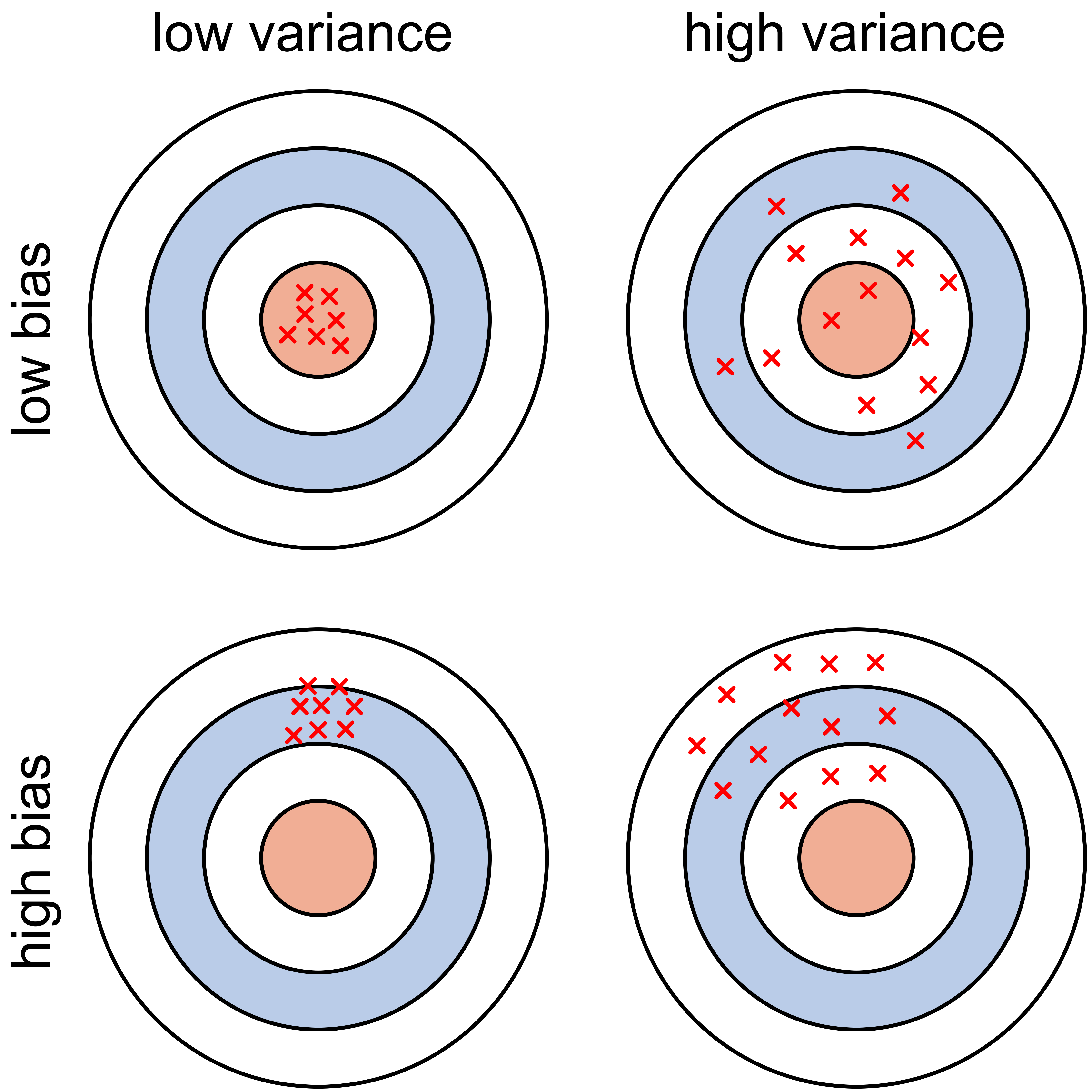
Training Data

Unknown Data

Source White Pony: Wikimedia Commons, CC BY-SA.
Source Brown Pony: Wikimedia Commons, CC BY-SA.
Source White Horse: Wikimedia commons, CC BY.

Bias and Variance

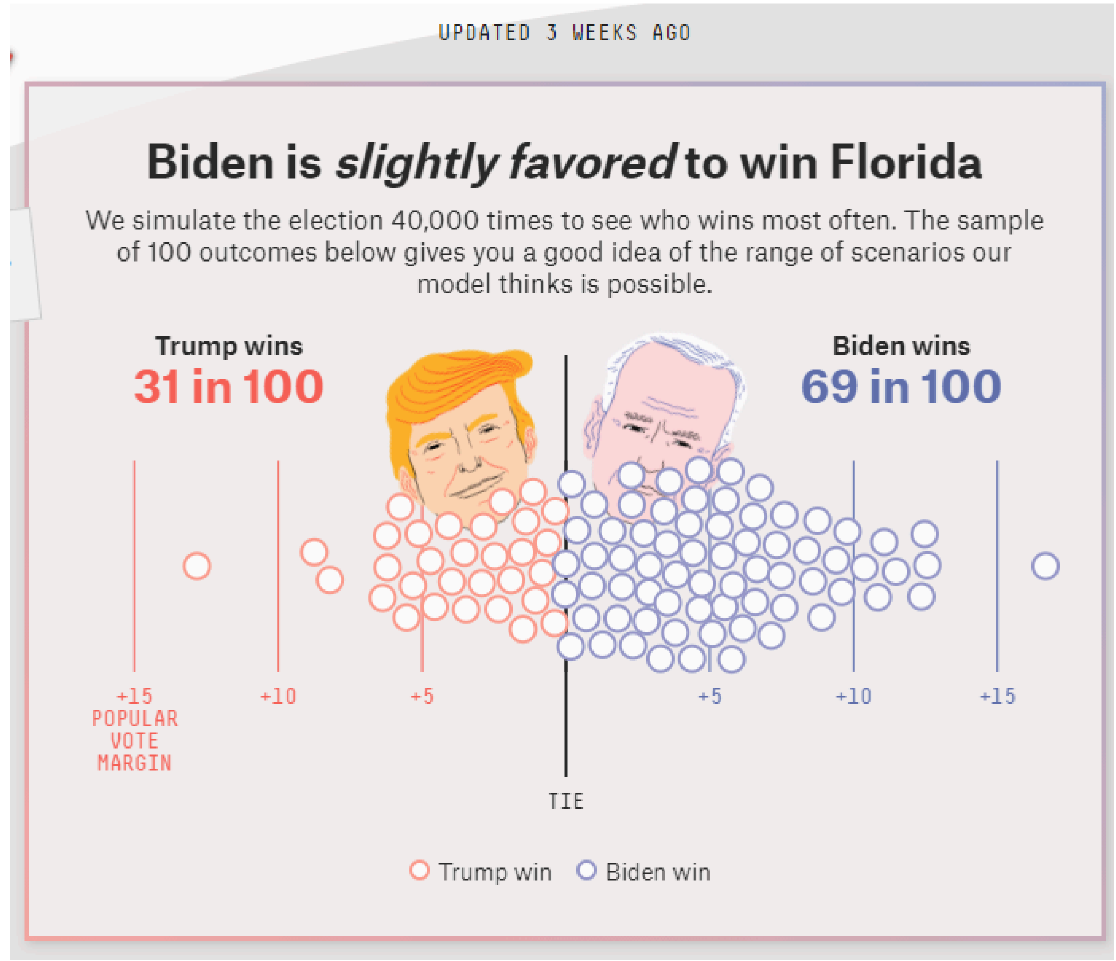
- In the context of machine learning, bias is also called **underfitting**:
- → the algorithm does not learn the relationships between attributes and outputs.
- Variance is also called **overfitting**:
- → the algorithm models false correlations or noise instead of the desired relationships between attributes and outputs.



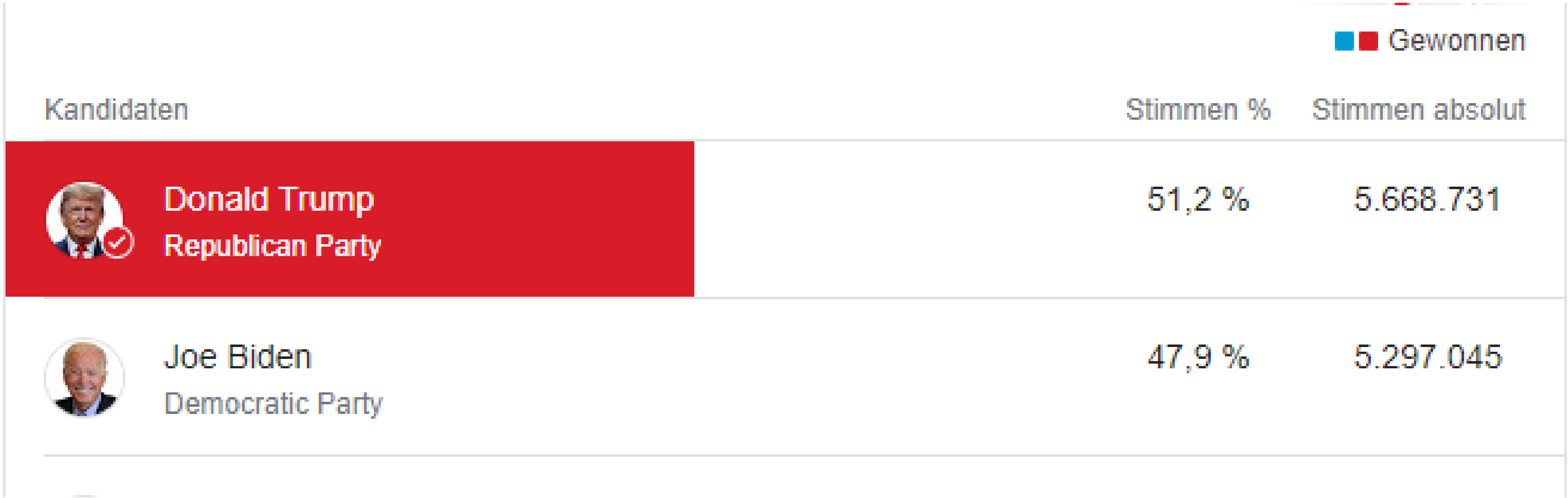
Source illustration: Adapted from Scott Fortmann-Roe Graphical illustration of bias and variance.

Example: Election Prediction

Prediction for Florida 2020



Actual Result



Gewonnen

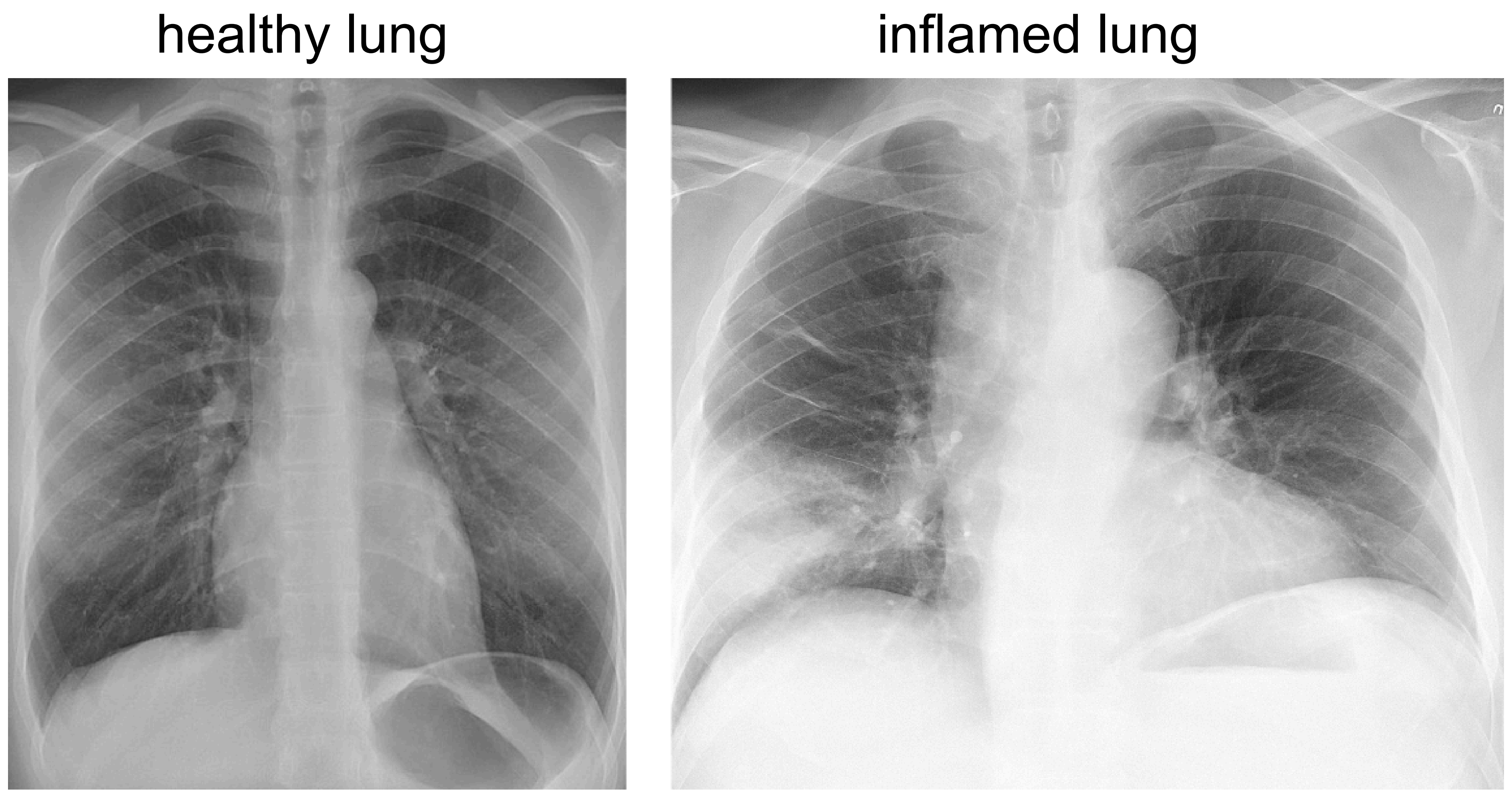
Kandidaten	Stimmen %	Stimmen absolut
Donald Trump Republican Party	51,2 %	5.668.731
Joe Biden Democratic Party	47,9 %	5.297.045

Possible Sources of Error

- **Bias**: Attributes of respondents such as education or landline connection were not considered → the prediction systematically underestimates the number of Trump voters.
- **Variance**: Not enough people surveyed, the model learns the attributes of individual people → the prediction is strongly influenced by the random selection of respondents.

Example: Pneumonia Prediction

- **Goal**: Detect pneumonia in X-ray images of lungs.
- **Training dataset:** X-ray images from three hospitals.
- Incidence of pneumonia:
  - Hospital 1: 1.2%
  - Hospital 2: 34.2%
  - Hospital 3: 1.0%.
- The machine learning model receives both the images and the **metadata** (which hospital, which device, time, etc.) for training.



X-ray images of lungs. Source: Wikimedia Commons, public domain.

Possible Problems?

*The model learns the **source hospital** of images, not how pneumonia looks like → **Bias**.*

See also Zech et al., PloS Medicine (2018).

Sources of Bias

- **Data Selection**: Datasets should represent the underlying population.

  Example: a facial recognition algorithm trained on images of predominantly white men that poorly recognizes black women.

- **Prejudices and Correlations**: Attributes can contain undesirable correlations that reflect prejudices.

  Example: a recruiting algorithm that includes postal code and thus discriminates against migrants.

- **Missing Information**: Attributes not included in the dataset can be very important for the result.

  Example: Predicting customer willingness to buy. Customers from country X spend twice as much money as customers from country Y, but the country of origin is not collected and is not available for prediction.

- **Different Data Source**: when the data source for the training data is different from the data used in the application.

  Example: the training data for an image recognition algorithm is captured with a Nikon camera, all data for the application comes from Canon cameras.

Correct Assessment of Model Error

In supervised learning, algorithms tend to "memorize" their training data.
→ the prediction error on the training data is not a good measure of variance!

Train-Test Approach

1. Divide the data with known classes into two parts: the **training dataset** (e.g., 80% of the data) and the **test dataset** (e.g., 20% of the data).
2. Train the model only with the training dataset.
3. Predict the classes in the test dataset with the trained model.
4. Compare the predicted classes with the true classes.
5. (Optional) Separate a third part (**validation data**) to evaluate the model during development.

Summary

- Errors of machine learning models have two reasons:
  - **Bias**: a systematic deviation of model predictions from the true value.
  - **Variance**: a high spread of predictions for similar input values.
- Sources of bias and variance can lie in both the training data and the training process.
- Systematic identification of biases in training data helps to reduce model bias.
- Splitting data into training and test datasets helps prevent overfitting and correctly assess model performance.